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PAASCU LEVEL II: Higher Education
(Philippine Accrediting Association of Schools, Colleges and Universities)

PROJECT PROGRESS REPORT

ARDUINO: SMART TRASHBIN

Group 3

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l. Project Overview

The Smart Trashbin Arduino Project is an automated waste disposal system designed to promote hygiene and convenience. The main objective of this project is to create a touchless trash bin that automatically opens its lid when a person's hand or object is detected nearby.

The system uses an Arduino microcontroller, an ultrasonic sensor to detect distance, and a servo motor to control the lid movement. By minimizing physical contact, the project helps reduce the spread of germs and encourages proper waste disposal in homes, schools, and public areas.

ll. Current Status

So far, the following components and features have been completed:

Hardware Setup

- Arduino board connected and powered properly
- Ultrasonic sensor wired for distance detection
- Servo motor connected for lid movement
- Breadboard and jumper wires organized correctly

Programming

- Basic Arduino code uploaded successfully
- Ultrasonic sensor can measure distance accurately
- Servo motor rotates when an object is detected within the set range
- Lid automatically closes after a few seconds

Initial Testing

- System responds to hand detection
- Opening and closing mechanism works
- Basic troubleshooting completed
- The trash bin is functional but still needs improvements in stability and design.

III. Challenges & Solutions

Challenge 1: Inconsistent Distance Detection

At first, the ultrasonic sensor gave unstable readings.

Solution:

We adjusted the sensor angle and added slight delays in the code to stabilize the readings.

Challenge 2: Servo Motor Not Moving Properly

The servo motor sometimes did not rotate smoothly.

Solution:

We checked the wiring connections and ensured proper power supply. We also adjusted the servo angles in the code to avoid over-rotation.

Challenge 3: Jumper M-to-M Wire Connection Issues

We experienced loose connections using male-to-male (M-to-M) jumper wires, which caused the system to randomly stop working or the servo to malfunction.

Solution:

We secured the jumper wires firmly into the breadboard and Arduino pins. We also replaced some worn-out jumper wires and rearranged the wiring layout to prevent accidental disconnections.

To complete the project, the following tasks remain:

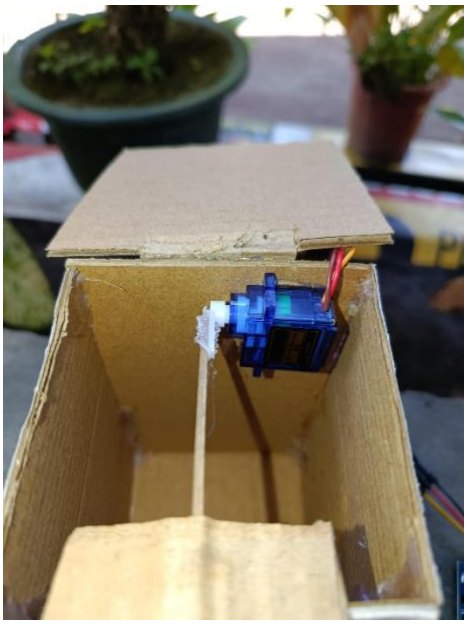
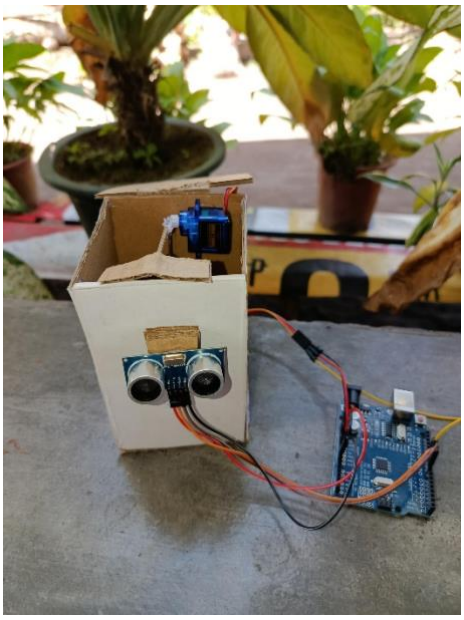
- Improve wiring for cleaner presentation
- Add a protective casing for the Arduino and components
- Test the system for long-term durability

IV. Future Improvements:

LCD display for status messages

Buzzer sound when the lid opens

Trash level detection sensor

Once these improvements are completed, the Smart Trashbin will be fully functional, reliable, and ready for demonstration.

Documentation: